WORKED SOLUTION

STEP I 2011, Question 13

Probability and Statistics • Continuous Random Variables

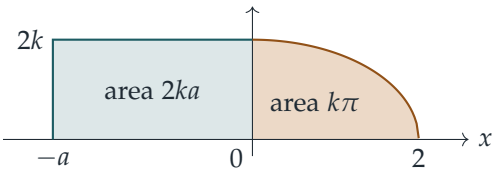
STEP I 2011, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q13

The key idea

Draw the density. It is a rectangle of height $2k$ over $[-a, 0)$ joined to a quarter of a circle of radius 2 (scaled vertically by k) over $[0, 2]$. Areas under it are probabilities, so the rectangle and the quarter-circle do most of the work without any integration. The total area must be 1, and that fixes k. After that, every question about d comes down to asking how much area lies to the right of a point.



Solution

Finding k

On $[0, 2]$ the graph of $\sqrt{4 - x^2}$ is a quarter of the circle $x^2 + y^2 = 4$, so

$$\int_0^2 \sqrt{4 - x^2}\,\mathrm{d}x = \tfrac{1}{4}\pi \cdot 2^2 = \pi.$$

(The given antiderivative agrees: $2\arcsin 1 + 0 - 0 = \pi$.) The total probability is therefore $2ka + k\pi = 1$, so

$$k = \frac{1}{2a + \pi}.$$

Part (i)

We split the integral for $\mathrm{E}(X)$ at 0:

$$\int_{-a}^0 2kx\,\mathrm{d}x = k[x^2]_{-a}^0 = -ka^2.$$

For the second piece, the derivative of $(4-x^2)^{3/2}$ is $-3x(4-x^2)^{1/2}$, so

$$\int_0^2 kx\sqrt{4-x^2}\,\mathrm{d}x = k\Big[-\tfrac{1}{3}(4-x^2)^{3/2}\Big]_0^2 = k\cdot\tfrac{1}{3}\cdot 8 = \tfrac{8}{3}k.$$

Adding the two pieces,

$$\mathrm{E}(X) = k\Big(\frac{8}{3}-a^2\Big) = \frac{8-3a^2}{3(2a+\pi)}.$$

Part (ii)

The probability to the right of 0 is the quarter-circle area:

$$\mathrm{P}(X>0) = k\pi = \frac{\pi}{2a+\pi}.$$

Ask yourself How does knowing $\mathrm{P}(X>0)$ tell us the sign of d?

The function $t \mapsto \mathrm{P}(X>t)$ is continuous and strictly decreasing for $-a \le t \le 2$, because the density is positive on $(-a,2)$. It falls from 1 to 0, so exactly one d satisfies $\mathrm{P}(X>d) = \frac{1}{10}$. Since the function is decreasing, $d<0$ exactly when $\mathrm{P}(X>0) < \frac{1}{10}$. In other words, d is negative when less than a tenth of the probability lies to the right of 0.

Now suppose $2a > 9\pi$. Then $2a+\pi > 10\pi$, so

$$\mathrm{P}(X>0) = \frac{\pi}{2a+\pi} < \frac{1}{10},$$

and hence $d<0$, as required. (The argument also runs backwards: $d<0$ only if $2a>9\pi$.)

In this case $-a \le d < 0$, so the area to the right of d is the part of the rectangle from d to 0, of width $-d$, together with the whole quarter-circle:

$$\mathrm{P}(X>d) = 2k(-d) + k\pi = \frac{\pi-2d}{2a+\pi}.$$

Setting this equal to $\frac{1}{10}$ gives $10\pi - 20d = 2a+\pi$, so

$$d = \frac{9\pi-2a}{20}.$$

This value really does lie in $[-a,0)$: it is negative because $2a>9\pi$, and $d \ge -a$ is equivalent to $9\pi+18a \ge 0$, which is true.

Part (iii)

Now $d=\sqrt{2}>0$, so d lies under the circular part and

$$\mathrm{P}(X>\sqrt{2}) = k\int_{\sqrt{2}}^2 \sqrt{4-x^2}\,\mathrm{d}x.$$

Using the given result,

$$\begin{aligned}\int_{\sqrt{2}}^{2} \sqrt{4-x^2}\,\mathrm{d}x &= \left[2\arcsin(\tfrac{1}{2}x) + \tfrac{1}{2}x\sqrt{4-x^2}\right]_{\sqrt{2}}^{2} \\ &= (2\cdot\tfrac{\pi}{2}+0) - (2\cdot\tfrac{\pi}{4} + \tfrac{\sqrt{2}}{2}\cdot\sqrt{2}) \\ &= \frac{\pi}{2} - 1.\end{aligned}$$

(Geometrically, this is the circular segment cut off by the chord at $x=\sqrt{2}$. It equals an eighth of the disc, $\frac{\pi}{2}$, minus a right-angled triangle of area 1.) So

$$\frac{\frac{\pi}{2}-1}{2a+\pi} = \frac{1}{10} \quad\Longrightarrow\quad 5\pi - 10 = 2a+\pi \quad\Longrightarrow\quad a = 2\pi - 5.$$

This is positive ($a \approx 1.283$), and it satisfies $2a \le 9\pi$, which agrees with part (ii) since $d > 0$.

Checks

- In part (i), with $a=1$ the mean is $\frac{5}{3(2+\pi)} \approx 0.3242$. A Monte Carlo simulation with 2×10^5 samples gave 0.3236.
- The formula in part (ii) gives $d=0$ at the borderline $2a = 9\pi$, which is consistent with the condition. With $a=20$ it gives $d = \frac{9\pi-40}{20} \approx -0.586$, and solving $\mathrm{P}(X > d) = \frac{1}{10}$ numerically gives the same value.
- In part (iii), with $a = 2\pi - 5$, Simpson's rule gives $\mathrm{P}(X > \sqrt{2}) = 0.1000$.

Common slips

- Leaving k unknown. The normalisation $2ka + k\pi = 1$ is the hidden first step, and every answer is in terms of a alone.
- In part (ii), forgetting the quarter-circle and writing $\mathrm{P}(X > d) = -2kd$ only. All of the probability on $[0,2]$ also lies to the right of a negative d.
- Getting the sign of $\int_{-a}^{0} 2kx\,\mathrm{d}x$ wrong. It is negative, because $x < 0$ throughout.

Beyond STEP

The value d is the upper 10% point, or 0.9-quantile, of X. Finding it means inverting the cumulative distribution function F, and that inverse is exactly what is used to simulate a random variable from a uniform one (inverse transform sampling). The mixed shape of this density, flat on one side and curved on the other, shows why quantiles often have to be found piece by piece. Grimmett and Welsh, *Probability: An Introduction*, covers distribution functions and quantiles carefully.

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